Rashtreeya Sikshana Samithi Trust

RV COLLEGE OF ENGINEERING®

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF MATHEMATICS

STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS (MA231TB)

(Common to AS, BT, CH, IM, ME)

MANUAL FOR EXPERIENTIAL LEARNING USING MATLAB

III – SEMESTER

Contents

Introduction

The students are exposed to the good learning of coding using MATLAB in I year and have got introduced to the coding part for the topics in the Engineering Mathematics syllabus of I and II semesters.

“The learning never stops, the more the better”.

The department of Mathematics has taken all efforts to give best knowledge of Mathematics relevant to the programmes. In this direction the syllabus is framed in a branch specific way enabling the different programmes to take this as basis and develop the further theoretical knowledge.

To make the students more industry-ready, the MATLAB coding is included in the syllabus and the students are made to be aware of specific MATLAB functions.

This manual is made-to-order MATLAB content for the specific syllabus and numbers of application problems are included.

After going through the manual, the students would be well equipped to solve any such related problems.

1 Statistics

Topic learning outcomes:

Student will be able to:

1. Find measures of central tendency: moments, skewness and kurtosis.
2. Fit data into several types of curves.
3. Obtain correlation coefficient and regression lines.

Syntax and description:

- m = moment(X,order) returns the central moment of X for the order specified by order.
- y = skewness(X)returns the sample skewness of X.
- k = kurtosis(X) returns the sample kurtosis of X.
- R = corrcoef(A,B)returns coefficients between two variables A and B.
- fitobject = fit(x,y,fitType)creates the fit to the data in x and y with the model specified by fitType.
- R = corrcoef(A,B) returns correlation coefficient between two random variables A and B.
- [r,m,b] = regression(x,y) returns correlation coefficient 'r' between two variables x and y, slope 'm' of the regression line of y on x, i.e., y= mx+ b and constant term 'b'.

Note: fitType can be

1. Linear model 'Poly1':
   f(x) = p1*x + p2
2. Quadratic model 'Poly2':
   f(x) = p1*x^2 + p2*x + p3
3. General model 'Exp1':
   f(x) = a*exp(b*x)
4. General model 'Power1': f(x) =
   a*x^b

Example 3.1 Wages of workers in a company are given in the following table:

Wages	50-70	70-90	90-110	110-130	130-150	150-170	170-190
No. of workers(f)	4	8	12	20	6	7	3

Calculate the first four central moments, skewness and kurtosis for the following distribution.

Note : 'x' is taken as the middle value of class interval, 'f' is no. of times x is repeated.

```
x=[60   60  60 80 80 80      80 80 80 80      80 100 100 100 100 100 100 100 100 100 100
60
100 100 120      120 120 120      120 120 120      120 120 120 120 120 120 120 120 120 120
120 120 120 140 140 140          140 140 140      160 160 160 160 160 160 160 180 180 180];
y1=moment(x,1)
```

```
y2=moment(x,2)
y3=moment(x,3)
y4=moment(x,4) sk =
skewness(x) k =
kurtosis(x)
```

ans = y1
= 0

y2 = 926.5556 y3 =
4.7747e+03 y4 =
2.1951e+06 sk =
0.1693

k = 2.5569

Example 3.2: Fit a straight line to the following data.

x	1	6	11	16	20	26
y	13	16	17	23	24	31

```
x = [16      11      16      20      26]';
y =[ 13  16      17  23      24      31]';

f=fit(x,y,'Poly1') plot(f,x,y)
```

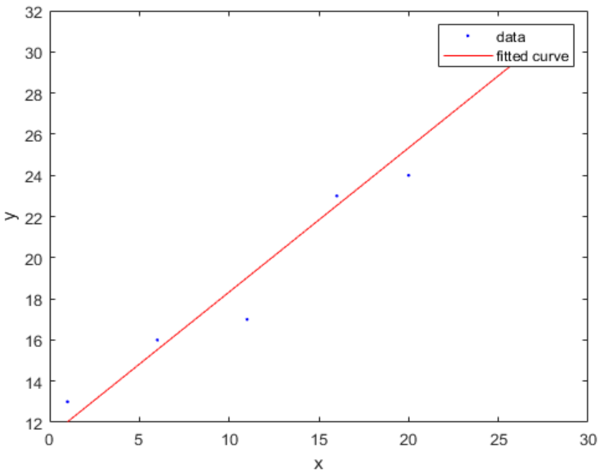
f =

Linear model Poly1:

f(x) = p1*x + p2

Coefficients (with 95% confidence bounds): p1 = 0.7008
(0.4936, 0.908)

p2 = 11.32 (8.057, 14.59)



Example 3.3: The following table gives the results of the measurements of train resistances; V is the velocity in mile per hour and R is the resistance in pound per ton.

V	20	40	60	80	100	120
R	5.5	9.1	14.9	22.8	33.3	46

If R is related to V by the relation $R= a + bV + cV^2$

```
x=[20  40    60    80    100    120]';
y=[5.5 9.1   14.9  22.8  33.3   46]';
f=fit(x,y,'Poly2')
plot(f,x,y)
```

f =

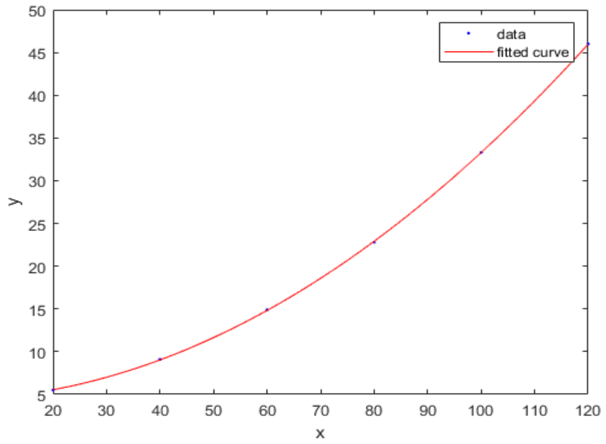
Linear model Poly2:

f(x) = p1*x^2 + p2*x + p3

Coefficients (with 95% confidence bounds): p1 = 0.002871 (0.002753, 0.002989)

p2 = 0.002411 (-0.01447, 0.01929)

p3 = 4.35 (3.834, 4.866)



Example 3.6 The following table gives the stopping distance y in meters of a motor bike moving at a speed of x kms/hour when the breaks are applied.

x	16	24	32	40	48	56
y	0.39	0.75	1.23	1.91	2.77	3.81

Find the correlation coefficient between the speed and the stopping distance, and the equations of regression lines. Hence estimate the maximum speed at which the motor bike could be driven if the stopping distance is not to exceed 5 meters.

```
x=[16  24     32     40     48     56]';
y=[0.39       0.75   1.23   1.91   2.77   3.81]';
R = corrcoef(x,y)
[r,m,b] = regression(x,y)
```

```
ans   =
R =   2×2
      1.0000       0.9827
          0.98     1.0000
          27
    ]     0.98
          27
    :
    ]     0.08
          51
    :
    ]     -
          1.255
    :     1
```

Exercise:

1. Find the second, third and fourth central moments of the frequency distribution given below. Hence, find (i) a measure of skewness and (ii) a measure of kurtosis.

Class limits	110.0 – 114.9	115.0 – 119.9	120.0 – 124.9	125.0 – 129.9	130.0 – 134.9	135.0 – 134.9	140.0 – 144.9
Frequency	5	15	20	35	10	10	5

2. Find the second, third and fourth central moments of the frequency distribution given below. Hence, find (i) a measure of skewness and (ii) a measure of kurtosis.

x	5	10	15	20	25	30	35
f	4	3	2	8	10	10	5

3. The latent heat of vaporisation of steam r is given in the following table at different temperature t:

t	40	50	60	70	80	90	100	110
r	1069.1	1063.6	1058.2	1052.7	1049.3	1041.8	1036.3	1030.8

Fit a relation to the form r =a+bt.

4. The following table gives the result of the measurement of train resistances; V is the velocity in miles per hour; R is the resistance in pounds per ton:

V	20	40	60	80	100	120
R	5.5	9.1	14.9	22.8	33.3	46.0

Fit a relation to the form $R = a + bV + cV^2$.

5. The following data represent carbon dioxide (co_2) emission from coal-fluid boilers (In units of 1000 tons) over a period of years 2005 to 2017. The variable (year) has been standardized and yield following table.

Years(x)	0	5	8	9	10	11	12
co_2 emission (y)	910	680	520	450	370	350	340

Find regression model of y on x.

6. Obtain the lines of regression and hence find the coefficient of correlation for the following data

X	1	3	4	2	5	8	9	10	13	15
Y	8	6	10	8	12	16	16	10	32	32

2. Complex Integration

Topic learning outcomes:

Student will be able to:
1. Evaluate line integral and contour integral in the complex plane
2. Verify Cauchy's theorem

Syntax and description:

- q = integral(fun,zmin,zmax)- integrates complex function fun from zmin to zmax.
- q = integral(fun,zmin,zmax,Name,Value) specifies additional options with one or more Name,Value pair arguments. For example, specify 'WayPoints'followed by a vector of real or complex numbers to indicate specific points for the integrator to use.

Example 2.1: Compute $\int_c |z|^2 dz$ around the square with vertices at (0,0), (1,0), (1,1), (0,1).

```
fun = @(z) abs(z.^2);
C = [1 1+i i];
p = integral(fun,0,0,'Waypoints',C)
```

```
p =
  -1.0000 + 1.0000i

p = -9.8696
```

Example 2.2: Obtain $\int_{1-i}^{2+3i} (z^2 + z)dz$ along the line joining the points A(1,-1) and B(2,3).

```
fun = @(z) z.^2+z;
%C = [-i i];
p = integral(fun,1-i,2+3*i)
```

```
p =
 -17.1667 +10.6667i
```

Example 2.3: Evaluate $\int_c (z - z^2)dz$ where c is the upper half of the circle |z-2|=3.

$$z = 2 + 3e^{i\theta}$$

```
fun = @(theta) -(2+9.*exp(i*theta)+9.*exp(i*2*theta)).*(i*3*exp(i*theta));
%C = [1 1+i i];
p = integral(fun,0,pi)
```

```
p =
  30.0000 - 0.0000i
```

Example 2.4: Find $\int_c \frac{e^{3z}}{(z-\log 2)^4} dz$ where c is the square with vertices $\pm 1 \pm i$.

```
fun = @(z) exp(3*z)./(z-log(2)).^4;
C = [1-i 1+i -1+i];
p = integral(fun,-1-i,-1-i,'Waypoints',C)
```

```
p =
   1.8774e-14 + 2.2619e+02i
```

Exercise:

1. Find $\int_0^{1+i\pi} (z^2 + \cosh 2z) dz$.
2. Integrate the function $\frac{z+2}{z}$ along the curve c: circle of radius 2, centre origin.
3. Verify Cauchy's theorem for the integral of $f(z) = \frac{1}{z}$ taken over the triangle formed by the points (1,2), (3,2) and (1,4).
4. Evaluate $\int_{1-i}^{2+3i} (z^2 + z) dz$ along the line joining the points A(1,-1) and B(2,3).

3 Laplace Transform

Topic learning outcomes:

Student will be able to:
1. Determine the Laplace transform of elementary functions.
2. Develop the Laplace transform of periodic function, Heaviside (unit step) function and Dirac Delta function.

Syntax and description:

- laplace(f)- returns the Laplace transform of function f. By default, the independent variable is t and transformation variable is s.
- laplace(f,transVar)- uses the transformation variable transVar instead of s.
- laplace(f,var,transVar)- uses the specified independent variable var and the transformation variable transVar instead of t and s respectively.

Example 1.1: Compute the Laplace transform of the signal $f(t) = t + t^{\frac{1}{4}} + t^{\frac{9}{2}} + e^{-bt} + \sin\, \omega t - \cos nt$.

```
syms s t b w n
laplace(t +t^(1/4)+ t^(9/2) +exp(-b*t) +sin(w*t)-cos(n*t))
```

ans =

w/(s^2 + w^2) - s/(n^2 + s^2) + 1/(b + s) + 1/s^2 + (945*pi^(1/2))/(32*s^(11/2)) + (pi*2^(1/2))/(4*s^(5/4)*gamma(3/4))

Example 1.2: Obtain the Laplace transform of e^{-at} by specifying

(i) the transformation variable as *y*.

(ii) both the independent and transformation variables as *a* and *y* in the second and third arguments respectively.

(i)

```
syms a t y
f = exp(-a*t);
laplace(f, y)
```

ans =

$\frac{1}{a+y}$

(ii)

```
syms a t y
f = exp(-a*t);
laplace(f, a, y)
```

ans =

$\frac{1}{t+y}$

Example 1.3: Compute the Laplace transform of $f(t) = te^{-at} + \frac{coswt - cosnt}{t} + tsinnt$.

```
syms t a w n s
laplace(t*exp(-a*t)+(cos(w*t)-cos(n*t))/t +t*sin(n*t), t, s)
```

ans =

log(n^2/s^2 + 1)/2 - log(w^2/s^2 + 1)/2 + 1/(a + s)^2 + (2*n*s)/(n^2 + s^2)^2

Example 1.4: Find the Laplace transform of the function $f(t) = \delta(t-3) + H(t-\pi)$, where $\delta(t-3)$ is the Dirac delta function and $H(t-\pi)$ is the Heaviside function.

```
syms t s
laplace(dirac(t-3)+heaviside(t-pi), t, s)
```

ans =

exp(-3*s) + exp(-pi*s)/s

Example 1.5: Calculate $L[e^{t-2} + sin(t-2)]\, H(t-2)]$.

```
syms t s
f=(exp(t-2)+sin(t-2))*(heaviside(t-2))
laplace(f, t, s)
```

ans =

exp(-2*s)*(1/(s - 1) + 1/(s^2 + 1)

Example 1.6: Obtain the Laplace transform of $\int_0^t e^{-t} sint\ cost$.

```
syms t s
I = int(exp(-t)*sin(t)*cos(t),0,t)
simplify(laplace(I, t, s))
```

ans =

1/(s*(s^2 + 2*s + 5))

Example 1.7: Show that the Laplace transform of the derivative of a function is expressed in terms of the Laplace transform of the function itself.

```
syms f(t) s
Df = diff(f(t),t);
laplace(Df, t, s)
```

ans =

s*laplace(f(t), t, s) - f(0)

Exercise:

Compute the followings:

1. $L\left\{\sin(2t)+\int_0^t e^{-t}\cos t\,dt+2t^3\right\}$
2. $L\{sint\ sin2t\ sin3t + sinh2t + te^{-2t}\}$
3. $L[e^{t-2} + \sin(t-2)]\,H(t-2)]$
4. $L(e^{-4t}t^{-5/2})$
5. $L\left(\frac{e^{-t}\sin t}{t}\right)$
6. $L(\cosh^2 2t)$

4 Inverse Laplace Transform

Topic learning outcomes:

Student will be able to:
1. Evaluate the inverse Laplace Transform of the function.
2. Solve ordinary differential equations governed in real world situations.

Syntax and description:

- ilaplace(F)-returns the inverse Laplace transform of F. By default, the independent variable is s and the transformation variable is t. If F does not contain s, ilaplace uses the function symvar.
- ilaplace(F,transVar)-uses the transformation variable transVar instead of t.
- ilaplace(F,var,transVar)-uses the independent variable and transformation variable transVar instead of s and t respectively.

Example 2.1: Obtain the inverse Laplace transform of $\frac{1}{s^2}$.

```
syms s
F = 1/s^2;
ilaplace(F)
```

ans =
 t

Example 2.2: Compute the inverse Laplace transform of $\frac{1}{(s-a)^2}$. Also specify

(i) the transformation variable as *x*.

(ii) both the independent and transformation variables as *a* and *x*.

```
syms a s
F = 1/(s-a)^2;
ilaplace(F)
```

ans =
 t*exp(a*t)

(i)

```
syms x
ilaplace(F, x)
```

ans =
x*exp(a*x)

(ii)

```
syms x
ilaplace(F,a,x)
```

ans =
 x*exp(s*x)

Example 2.3: Determine the time domain signal from the corresponding frequency domain signal $F(s) = \frac{s}{s^2+4} + \frac{w}{s^2+w^2} + \frac{s}{s^2-n^2}$.

```
syms s w n
a = ilaplace(s/(s^2+4))
b = ilaplace(w/(s^2 + w^2))
c = ilaplace(s/(s^2 - n^2))
answer = a+b+c
```

```
a =

   cos(2*t)

b =

     sin(t*w)

c =

     exp(n*t)/2 + exp(-n*t)/2

answer =

    cos(2*t) + exp(n*t)/2 + exp(-n*t)/2 + sin(t*w)
```

Example 2.4: Compute the inverse Laplace transform of the following function $1 + \frac{e^{-2s}}{s^2+1} + tan^{-1}\left(\frac{1}{s}\right)$.

```
syms s t
F = 1+exp(-2*s)/(s^2+1)+atan(1/s);
ilaplace(F, s, t)
```

```
ans =

dirac(t) + sin(t)/t + heaviside(t - 2)*sin(t - 2)
```

Differential Equations using Laplace Transform:

Example 2.5: Solve the equation $y'' - 3y' + 2y = 4e^{2t}$ where $y(0) = -3, y(0) = 5$.

```
syms s t Y
f = 4 * exp(2 * t)
F = laplace(f,t,s)
Y1 = s * Y + 3
Y2 = s * Y1 - 5
Sol = solve(Y2 - 3 * Y1 + 2 * Y - F, Y)
sol = ilaplace(Sol,s,t)
```

```
f =
   4*exp(2*t)

F =
   4/(s - 2)

Y1 =
   Y*s + 3

Y2 =
   s*(Y*s + 3) – 5

Sol =
   (4/(s - 2) - 3*s + 14)/(s^2 - 3*s + 2)

sol =
   4*exp(2*t) - 7*exp(t) + 4*t*exp(2*t)
```

Example 2.6: Use the Laplace transform to determine the output of a system represented by the differential equation $y'' + y = x(t)$ in response to force function $H(t-1)$. Assume the initial conditions $y(0) = 0$, $y'(0) = 1$.

```
syms s t Y
f = heaviside(t-1)
F = laplace(f,t,s)
Y2 = s*s * Y - 1
Sol = solve (Y2 + Y - F, Y)
sol = ilaplace(Sol,s,t)
```

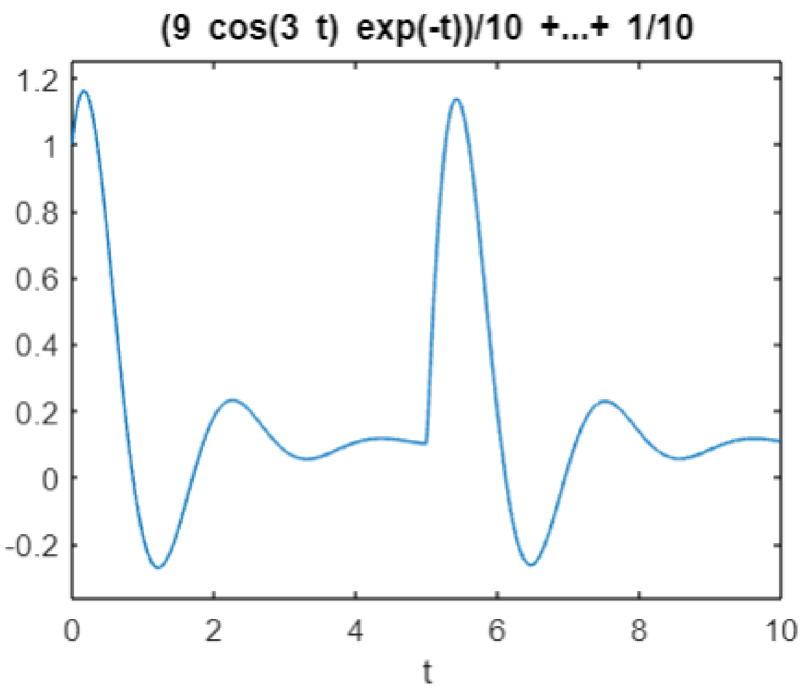
```
f =
  heaviside(t-1)

F =
   exp(-s)/s

Y2 =
   Y*s^2-1

Sol =
   (exp(-s)/s+1)/(s^2+1)

sol =
    sin(t)-heaviside(t-1)*(cos(t-1)-1)
```

Example 2.7: Solve the differential equation using Laplace transform
$y'' + 2y' + 10y = 1 + 5\,\delta(t-5)$, $y(0) = 1$, $y'(0) = 2$. Also plot the solution.

```
syms s t Y
f = 1 + 5*dirac(t-5)
F = laplace(f,t,s)
Y1 = s*Y - 1
```

```
Y2 = s*Y1 - 2
Sol = solve(Y2 + 2*Y1 + 10*Y - F, Y)
sol = simplify(ilaplace(Sol,s,t))
ezplot(sol,[0,10])
```

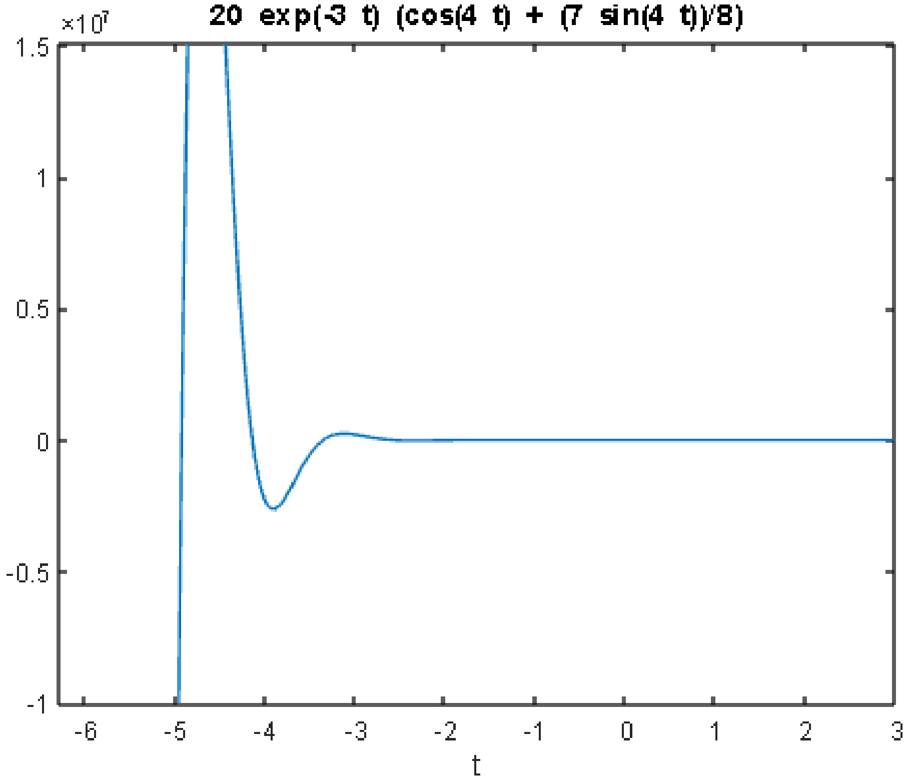
```
f =
  5*dirac(t - 5) + 1

F =
  5*exp(-5*s) + 1/s

Y1 =
  Y*s - 1

Y2 =
  s*(Y*s - 1) - 2

Sol =
 (s + 5*exp(-5*s) + 1/s + 4)/(s^2 + 2*s + 10)

sol =
 (9*cos(3*t)*exp(-t))/10 + (29*sin(3*t)*exp(-t))/30 +     (5*heaviside(t - 5)*exp(5 - t)*sin(3*t -
15))/3 + 1/10
```



Example 2.8: A particle is moving along a path satisfying the equation $\frac{d^2x}{dt} + 6\frac{dy}{dx} + 25x = 0$, where x denotes the displacement of the particle at time t. If the initial position of the particle is $x = 20$ and the initial speed is 10, find the displacement of the particle at any time t using Laplace transform.

```
syms s t X
a=20                    % initial displacement
b=10                    % initial velocity
X1=s*X-a                   % Laplace of dx/dt, X is Laplace of x(t)
X2 = s*s * X - s*a-b       % Laplace of second derivative of x(t)
Sol = solve(X2 +6*X1 +25*X, X)  % solution for X(s)
x(t) = ilaplace(Sol,s,t)       % solution for differential equation
ezplot(x(t))                 % graph of solution
```

```
a =
   20
b =
   10
X1 =
   X*s - 20
X2 =
   X*s^2 - 20*s - 10
Sol =
   (20*s + 130)/(s^2 + 6*s + 25)
x(t) =
   20*exp(-3*t)*(cos(4*t) + (7*sin(4*t))/8)
```



Example 2.9: A LCR circuit consists of a resistor R, a capacitor C and an inductor L connected in series together with a voltage source $e(t)$. Prior to closing the switch at time t = 0, both the charge on the capacitor and the resulting current in the circuit are zero. Determine the charge $q(t)$ on the capacitor in the circuit at time t given that R = 160 Ω, L = 1 H, C = 10^{-4} F and $e(t)$ = 20 V.

Solution: Applying Kirchhoff's second law to the circuit,

$$R\,i + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = e(t)$$

using $i = \frac{dq}{dt}$,

$$L\frac{d^2q}{dt^2}, +R\frac{dq}{dt} + \frac{1}{C}q = e(t)$$

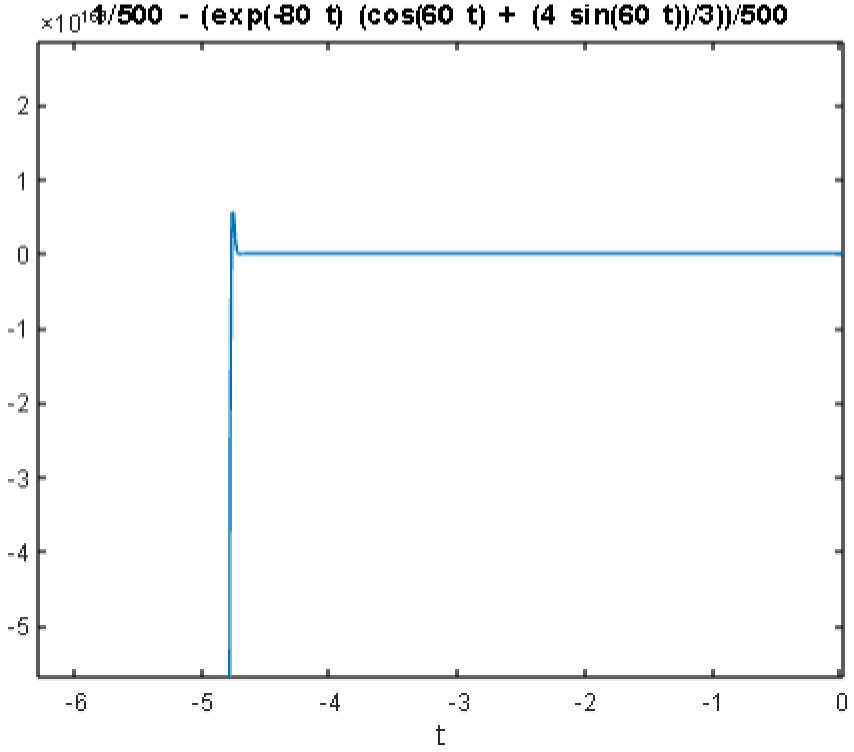
Substituting the given values for L, R, C and $e(t)$ gives

$$\frac{d^2q}{dt^2}, +160\frac{dq}{dt} + 10^4 q = 20$$

Solving this differential equation using MATLAB

```
syms s t Q
a=0;                    % initial displacement
b=0;                    % initial velocity
f=20                    % RHS of equation
F=laplace(f,t,s)           % Laplace of RHS
Q1=s*Q-a                % Laplace of dq/dt
Q2 = s*s * Q - s*a-b         % Laplace of second derivstive of q
Sol = solve(Q2 +160*Q1 +(10)^(4)*Q-F, Q)  % solution for Q(s)
q(t) = ilaplace(Sol,s,t)        % solution for differential equation
ezplot(q(t))                % graph of solution
```

```
f =
   20
F =
   20/s
Q1 =
   Q*s
Q2 =
   Q*s^2
Sol =
   20/(s*(s^2 + 160*s + 10000))
q(t) =
   1/500 - (exp(-80*t)*(cos(60*t) + (4*sin(60*t))/3))/500
```



Exercise:

1. Compute the following:

   (i) $L^{-1}\left\{\dfrac{4s+5}{(s-1)^2(s+2)}+\tan^{-1}\dfrac{a}{s}\right\}$

   (ii) $L^{-1}\left\{\dfrac{1}{(s^2+16)^2}+\dfrac{e^{-4s}}{(s-2)^4}\right\}$

   (iii) $L^{-1}\left\{\dfrac{s}{s^4+s^2+1}\right\}$

   (iv) $\mathrm{L}^{-1}\left\{\dfrac{2s^2+5s-4}{s^3+s^2-2s}\right\}$

2. An object undergoes forced vibrations according to the equation $\frac{d^2x}{dt^2}+16=\cos 2t$. If the particle starts from rest at $t=0$, find the displacement at any time $t>0$.

3. Solve $f'(t)=t+\int_0^t f(t-u)\cos u\,du,\ \ f(0)=4$ using Laplace transform.

4. Use the Laplace transform to find the output of the system described by the differential equation $y'''+2y''-y'-2y=x(t)$. In response to the input $x(t)=e^{-3t}$ and the initial conditions $y(0)=y'(0)=y''(0)=6$.

5 Numerical Methods

Topic learning outcomes:

Student will be able to:

1. Find the Numerical solutions to partial differential equations using finite difference method.
2. Solve the wave, heat and Laplace equation and other application problems.

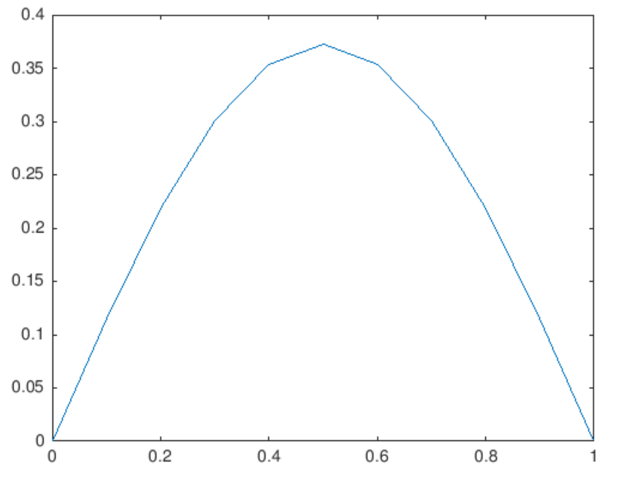
Note:

Using finite difference approximations to various derivatives, explicit finite difference schemes are obtained. These schemes are conditionally stable and require lot of iterations to converge to solution.

Example 5.1: Solve the one dimensional heat equation $u_t = u_{xx}$ subject to initial condition $u(x,0) = \sin(\pi x)$ and boundary conditions $u(0,t) = 0 = u(1,t)$ at $t = 0.1$, plot the solution.

MATLAB Program for heat equation

```
% Solution of 1D Heat equation using FD Scheme
% Length of rod, Time interval, Number of points in space
% Number of time steps, initial condition is u(x,0)=sin(pi*x)
% Boundaries are at zero temperature


L=1; T=0.1;k=1;N=10;M=50;
dx=L/N;
dt=T/M;
alpha=k*dt/dx^2;
for i=1:N+1
   x(i)=(i-1)*dx;
end
%Initial condition
for i=1:N+1
   uo(i)=sin(pi*x(i));
end
% Numerical scheme
for j=1:M
   for i=2:N
   u1(i)=uo(i)+alpha*(uo(i+1)-2*uo(i)+uo(i-1));
   end
   u1(1)=0;      % Boundary conditions
   u1(N+1)=0;    % Boundary conditions
   uo=u1;
end
%plot solution
plot(x,u1)
```

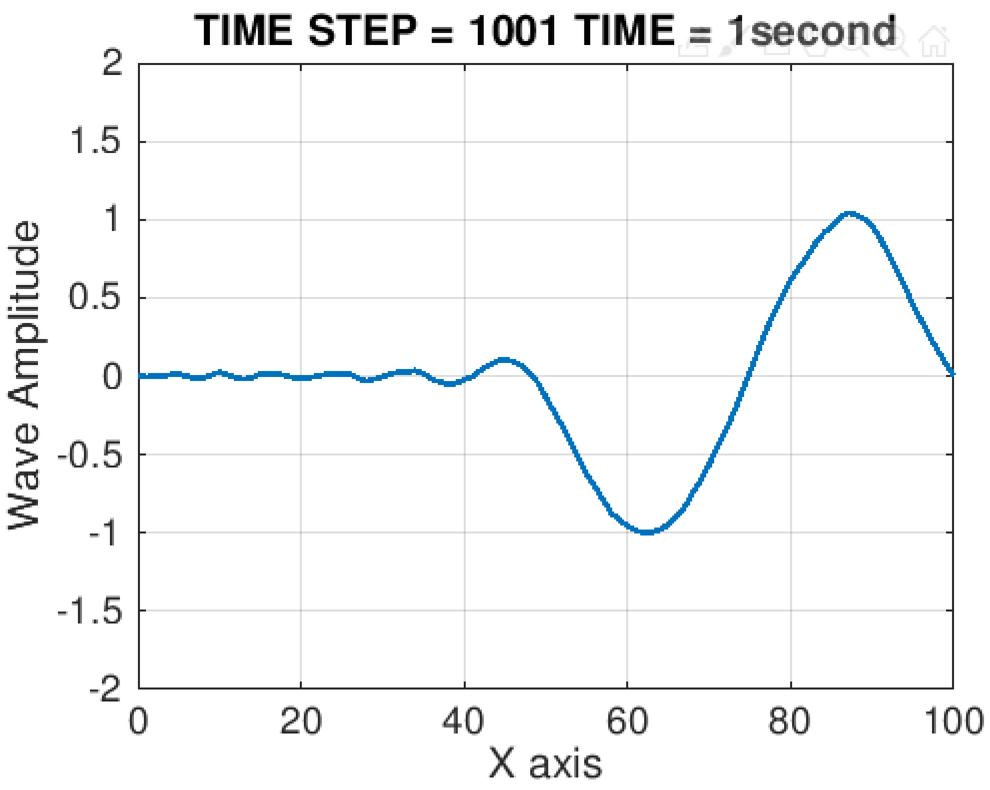


Example 5.2: Solve the one dimensional wave equation $u_{tt} = c^2 u_{xx}$ subject to initial conditions $u(x,0) = \sin(\pi x)$, $u_t(x,0) = 0$ and boundary conditions $u(0,t) = 0 = u(1,t)$, $c^2 = 500$, plot the solution.

MATLAB Program for wave equation

```
% This program describes a moving 1-D wave
% using the finite difference method
clc
close all;
clear all;
%---------------------------------------------------------------------%
% Initialization
Nx = 101; % x-Grids
dx = 1; % Step size
x(:,1) = (0:Nx-1)*dx;
mpx = (Nx+1)/2; % Mid point of x axis
% ( Mid pt of 1 to 101 = 51 here )
T = 1001; % Total number of time steps
f = 10; % frequency of source
dt = 0.001; % Time-Step
t(:,1)= (0:T-1)*dt;
v = 500; % Wave velocity
c = v*(dt/dx); % Convergence condition
U = zeros(T,Nx); % U(x,t) = U(space,time)
s1 = floor(T/f);
%---------------------------------------------------------------------%
% Initial condition
U((1:s1),1) = sin(2*pi*f.*t(1:s1));
U((1:s1),2) = sin(2*pi*f.*t(1:s1));
%---------------------------------------------------------------------%
% Finite Difference Scheme
for j = 3:T
```

```
for i = 2:Nx-1
U1 = 2*U(j-1,i)-U(j-2,i);
U2 = U(j-1,i-1)-2*U(j-1,i)+U(j-1,i+1);
U(j,i) = U1 + c*c.*U2;
end
end
%-------------------------------------------------------------------------%
% Plot the results
plot_times = [50 150 250 310];
for i = 1:4
figure(i)
k = plot_times(i);
plot(x,U(k,:),'linewidth',2);
grid on;
axis([min(x) max(x) -2 2]);
xlabel('X axis','fontSize',14);
ylabel('Wave Amplitude','fontSize',14);
titlestring = ['TIME STEP = ',num2str(k), ' TIME = ',num2str(t(k)), 'second'];
title(titlestring ,'fontsize',14);
h=gca;
get(h,'FontSize')
set(h,'FontSize',14);
fh = figure(i);
set(fh, 'color', 'white');
end
%-------------------------------------------------------------------------%
% The travelling wave
for j = 1:T
plot(x,U(j,:),'linewidth',2);
grid on;
axis([min(x) max(x) -2 2]);
xlabel('X axis','fontSize',14);
ylabel('Wave Amplitude','fontSize',14);
titlestring = ['TIME STEP = ',num2str(j), ' TIME = ',num2str(t(j)), 'second'];
title(titlestring ,'fontsize',14);
h=gca;
get(h,'FontSize')
set(h,'FontSize',14);
fh = figure(5);
set(fh, 'color', 'white');
F=getframe;
end
```

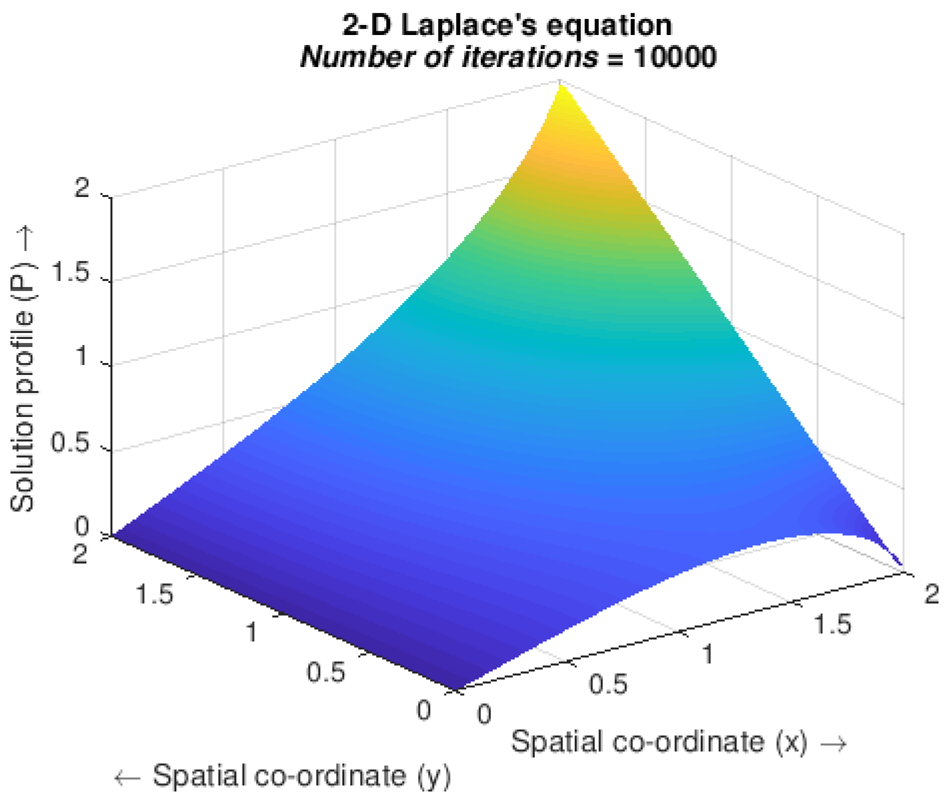


Example 5.3: Solve the two dimensional Laplace equation $u_{xx} + u_{yy} = 0$ subject to boundary conditions $x = 0\,, u = 0,\ \ x = 2\,,\ u = y\ ,\ y = 0\,, u = 0$ and $y = 2\,, u = 0$, plot the surface.

MATLAB Program for Laplace equation

```
% Solving the 2-D Laplace's equation by the Finite Difference
...Method
% Numerical scheme used is a second order central difference in space
...(5-point difference)
%%
%Specifying parameters
nx=60;                    %Number of steps in space(x)
ny=60;                    %Number of steps in space(y)
niter=10000;              %Number of iterations
dx=2/(nx-1);              %Width of space step(x)
dy=2/(ny-1);              %Width of space step(y)
x=0:dx:2;                 %Range of x(0,2) and specifying the grid points
y=0:dy:2;                 %Range of y(0,2) and specifying the grid points
%%
%Initial Conditions
p=zeros(ny,nx);           %Preallocating p
pn=zeros(ny,nx);          %Preallocating pn
%%
%Boundary conditions
p(:,1)=0;
p(:,nx)=y;
```

```
p(1,:)=p(2,:);
p(ny,:)=p(ny-1,:);
%%
%Explicit iterative scheme with C.D in space (5-point difference)
j=2:nx-1;
i=2:ny-1;
for it=1:niter
    pn=p;
    p(i,j)=((dy^2*(pn(i+1,j)+pn(i-1,j)))+(dx^2*(pn(i,j+1)+pn(i,j-1))))/(2*(dx^2+dy^2));
    %Boundary conditions
    p(:,1)=0;
    p(:,nx)=y;
    p(1,:)=p(2,:);
    p(ny,:)=p(ny-1,:);
end
%%
%Plotting the solution
surf(x,y,p,'EdgeColor','none');
shading interp
title({'2-D Laplace''s equation';['{\itNumber of iterations} = ',num2str(it)]})
xlabel('Spatial co-ordinate (x) \rightarrow')
ylabel('{\leftarrow} Spatial co-ordinate (y)')
zlabel('Solution profile (P) \rightarrow')
```



Exercise:

1. Find the solution of the parabolic equation $u_{xx} = 2u_t$ when $u(0,t) = 0 = u(4,t)$, taking h = 1 up to t = 5, plot the solution.

2. Solve the equation $u_{xx} = u_t$, subject to the condition $u(x,0) = \cos(\pi x)$, $0 < x < 1$ taking h = 1/3, k= 1/36.

3. Solve the equation $u_{xx} = u_{tt}$, $u(0,t) = 0 = u(1,t)$, $u(x,0) = \frac{1}{2}x(1-x)$, $u_t\,(x,0) = 0$, taking h = k= 0.1 for 0 < t < 0.4, plot the solution.

4. Solve the Laplace equation $u_{xx} + u_{yy} = 0$ subject to boundary conditions $x = 0\,, u = 0$, $x = 4\,,\ u = 12 + \ y$, $y = 0\,, u = 3x$ and $y = 4\,, u = x^2$, taking h = 1 = k, plot the surface.